Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Andrew Tkach	atkach@udel.edu
Friend (optional)	Troy Lankford	kavanagh@udel.edu

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The ter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is for people to have fun while playing the game. My project will be a simplified uno game.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are uno websites where you can play online or with friends.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want this to be used for people to have fun and enjoy themselves. Each route in my project can be what each card does and what happens if it gets played.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will scan each player's cards, see what happens when each card is placed and finish the game when a person runs out of cards.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to provide their name.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will compare input against the other databases and filter out options that don't work.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

They will get back a response if their card was able to be played or not.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Cards, Players(Users)

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Need: card color, card number/powerup, card amount Ignore: card size, types of players, player age

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the hardest part will be to code, shuffling the deck and giving people the right amount of cards with having a left over pile.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Using the LLM would be a great tool and I will ask the LLM questions until I understand what it is saying to me. If I still am stuck I will show it to the TA's in office hours and get help there.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

The goal is to put together a project with all the skills I learned during the course and do it in a fun way.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple drafter website on a simplified game of uno how could I do this?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

ChatGPT: If this is for your Python programming class, I'd work through these milestones: Milestone 1: Create the Card and State data classes. Milestone 2: Create a deck and deal five cards to each player. Milestone 3: Write and test can_play(). Milestone 4: Write the functions for playing and drawing cards. Milestone 5: Create the Drafter home page and display the player's hand. Milestone 6: Connect the buttons to the game logic. Milestone 7: Add the computer's turn, win conditions, and restart button. Milestone 8: Test the entire game, including cases where a player cannot play any cards.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

My Biggest Recommendation: Remove the Computer Opponent **This is probably the best simplification.** Making a computer opponent requires extra logic to search for valid cards and play them automatically. Instead, create a two-player game where players take turns using the same website.

For example:

1. Player 1 sees their cards and plays one.
2. The website displays a message saying it's Player 2's turn.
3. Player 2 plays a card.
4. The game continues until one player runs out of cards.

You would still need to hide Player 1's cards when Player 2 is taking their turn, but you wouldn't need any computer decision-making logic.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

It seems like it will be manageable and fun to create because there are simple solutions to these complex parts.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use the LLM and also office hours.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will go through each line in the code to try to debug it and if I still can't I will go to office hours for help.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top

of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

If this is for your Python programming class, I'd work through these milestones:
Milestone 1: Create the Card and State data classes.
Milestone 2: Create a deck and deal five cards to each player.
Milestone 3: Write and test can_play().
Milestone 4: Write the functions for playing and drawing cards.
Milestone 5: Create the Drafter home page and display the player's hand.
Milestone 6: Connect the buttons to the game logic.
Milestone 7: Add the computer's turn, win conditions, and restart button.
Milestone 8: Test the entire game, including cases where a player cannot play any cards.

My Biggest Recommendation: Remove the Computer Opponent

This is probably the best simplification.

Making a computer opponent requires extra logic to search for valid cards and play them automatically.

Instead, create a two-player game where players take turns using the same website.

For example:

5. Player 1 sees their cards and plays one.
6. The website displays a message saying it's Player 2's turn.
7. Player 2 plays a card.
8. The game continues until one player runs out of cards.

You would still need to hide Player 1's cards when Player 2 is taking their turn, but you wouldn't need any computer decision-making logic.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.